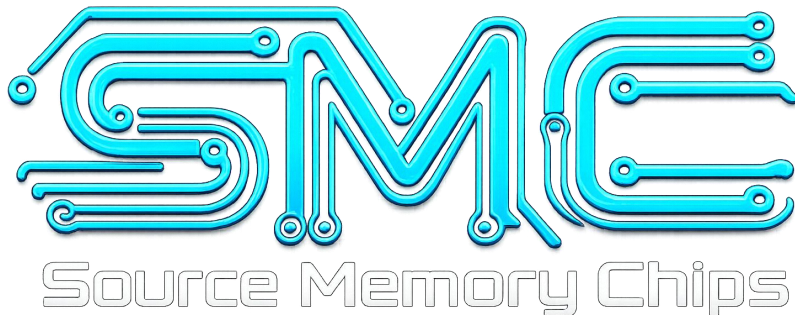


H9HP52ACPMAD

eMCP Memory Solution

Mobile / Embedded Memory Solution

Technical Datasheet



Source Memory Chips (SMC) provides technical reference documentation for memory products used in industrial, networking, embedded and long-life electronic applications.

<https://www.sourcememorychips.com>

SK hynix eMCP Memory Solution Product Specification



High Integration

64GB eMMC 5.1 + 32Gb LPDDR4X in a single eMCP package



High Speed

HS400 eMMC up to 400 MB/s; LPDDR4X up to 3733 Mbps



Compact Package

254-ball FBGA, 11.5 × 13.0 mm, 0.5 mm pitch



Mobile Ready

Designed for mobile and other space-constrained embedded platforms

1. Product Overview

The SK hynix H9HP52ACPMAD is an embedded multi-chip package (eMCP) memory solution integrating 64GB of eMMC 5.1 storage and 32Gb (4GB) of LPDDR4X SDRAM in a compact 254-ball FBGA package.

The single-package architecture combines managed non-volatile storage with high-speed system memory, reducing PCB area, simplifying signal routing and supporting space-constrained mobile and embedded platform designs.

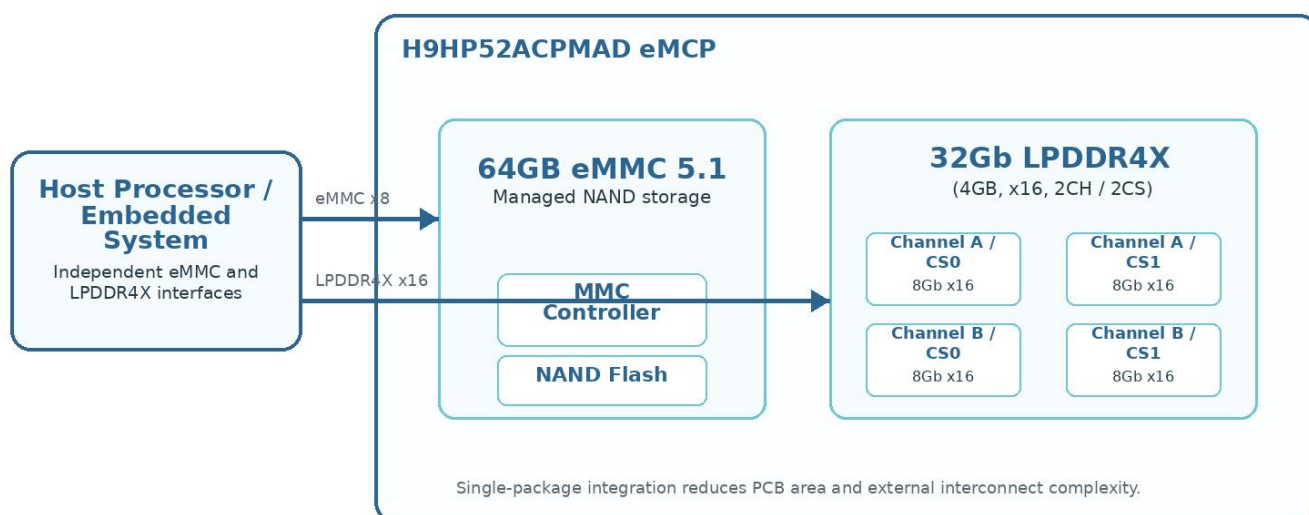
Item	Specification
Manufacturer	SK hynix
Content Model	H9HP52ACPMAD
Full Ordering Part	H9HP52ACPMADAR-KMM
Product Type	eMCP
Memory Configuration	64GB eMMC 5.1 + 32Gb (4GB) LPDDR4X
Package	254-ball FBGA
Operating Temperature	-25°C to +85°C

2. Memory Architecture

H9HP52ACPMAD integrates eMMC 5.1 storage and LPDDR4X working memory within one embedded package. The host accesses the storage subsystem through the eMMC interface and the DRAM subsystem through two independent LPDDR4X channels with two chip selects.

Memory Block	Function
64GB eMMC 5.1	Managed non-volatile storage for operating system, firmware, applications and user data.
32Gb LPDDR4X (4GB)	High-speed volatile memory for application processing and multitasking.

H9HP52ACPMAD Simplified Functional Architecture



3. Product Advantages

- Compact 64GB storage and 32Gb LPDDR4X integration in one package
- Reduced PCB footprint and external component count
- Simplified board routing and memory subsystem design
- HS400 eMMC support and LPDDR4X data rate up to 3733 Mbps
- Designed for mobile and other space-constrained embedded platforms

4. General Specifications

Parameter	Specification
eMMC Capacity	64GB, x8 data organization
eMMC Standard	JEDEC eMMC 5.1; backward compatible with eMMC 4.5 and 5.0
eMMC Bus Width	1-bit default, 4-bit and 8-bit modes
eMMC High-Speed Modes	HS400 and HS200; bus transfer rate up to 400 MB/s
LPDDR4X Capacity	32Gb, approximately 4GB
LPDDR4X Organization	x16, 2 channels / 2 chip selects
LPDDR4X Data Rate	Up to 3733 Mbps
Package	254-ball FBGA, 11.5 × 13.0 mm, 1.2 mm max
Environmental Status	Lead-free / Halogen-free / RoHS

Core Configuration

 64GB eMMC 5.1 Managed embedded storage	 32Gb LPDDR4X (4GB) x16, 2CH / 2CS	 3733 Mbps LPDDR4X Maximum data rate	 254-ball FBGA 11.5 × 13.0 mm package
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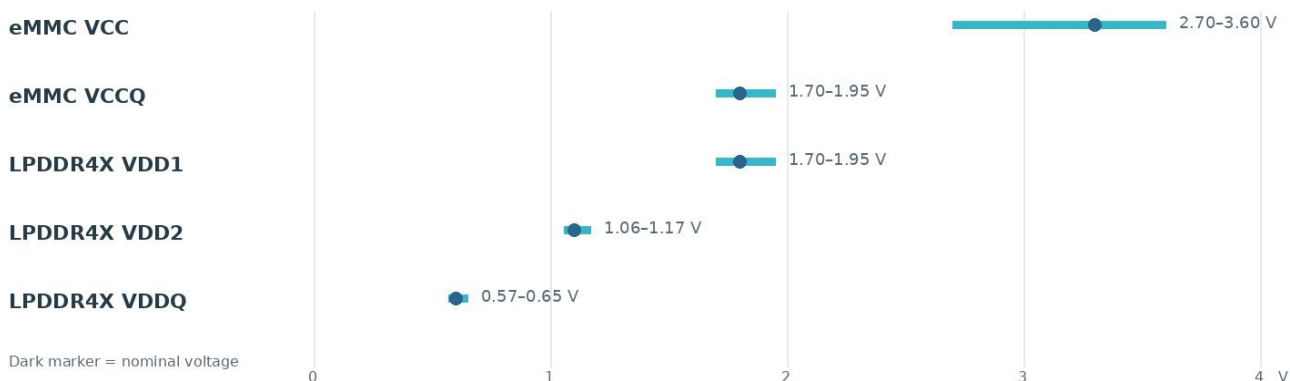
eMMC 5.1 Feature Support

Category	Supported Functions
Performance	HS400, HS200, enhanced data strobe and configurable driver strength
Data Management	Cache, cache flushing, discard, trim, erase and sanitize
Reliability	Reliable write, hardware/software reset, health report and field firmware update
Security / Partitioning	Partitioning, RPMB, write protection, lock/unlock and secure removal
Power Management	Power-off notification and sleep/awake support

5. Electrical Characteristics & Performance

Separate power rails are used for the eMMC subsystem and LPDDR4X memory. The ranges below summarize the operating windows specified by SK hynix; final power sequencing and decoupling should follow the official manufacturer specification.

Operating Voltage Range Overview



Performance Highlights

eMMC HS400

400 MB/s

Maximum bus transfer rate

LPDDR4X

3733 Mbps

Maximum data rate

eMMC Sequential Performance — Reference Device-Level Test

Read

280 MB/s

Write

170 MB/s

Conditions: HS400, 512KB sequential chunk, cache enabled; vendor reference values, not 100% tested.

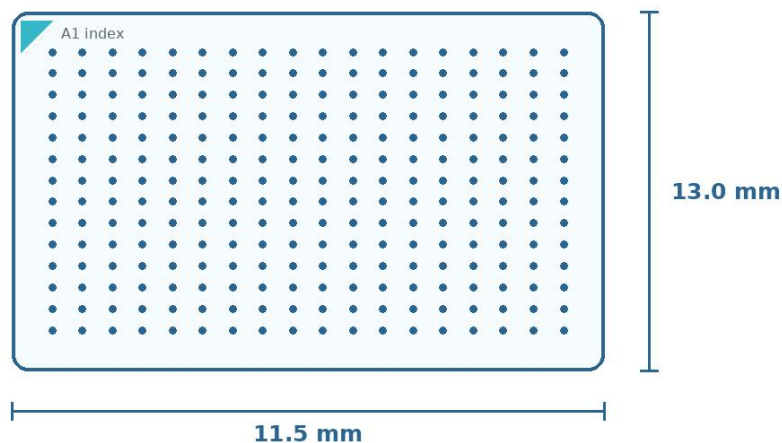
The 280 MB/s sequential read and 170 MB/s sequential write values are reference device-level results for the 64GB configuration under SK hynix test conditions (HS400, 512KB sequential chunk, cache enabled). Actual performance depends on host controller, firmware, workload and board design.

6. Package Information

H9HP52ACPMAD is associated with a compact 254-ball FBGA package measuring 11.5 mm × 13.0 mm with 0.5 mm ball pitch and 1.2 mm maximum package thickness.

254-ball FBGA Package — Simplified Mechanical View

- Ball pitch: 0.5 mm
- Maximum thickness: 1.2 mm
- Lead-free and halogen-free



Simplified illustration only — use the official SK hynix package drawing and ball map for PCB design.

Mechanical Item	Specification
Package Type	254-ball FBGA
Body Size	11.5 mm × 13.0 mm
Maximum Thickness	1.2 mm
Ball Pitch	0.5 mm
Package Material	Lead-free and halogen-free

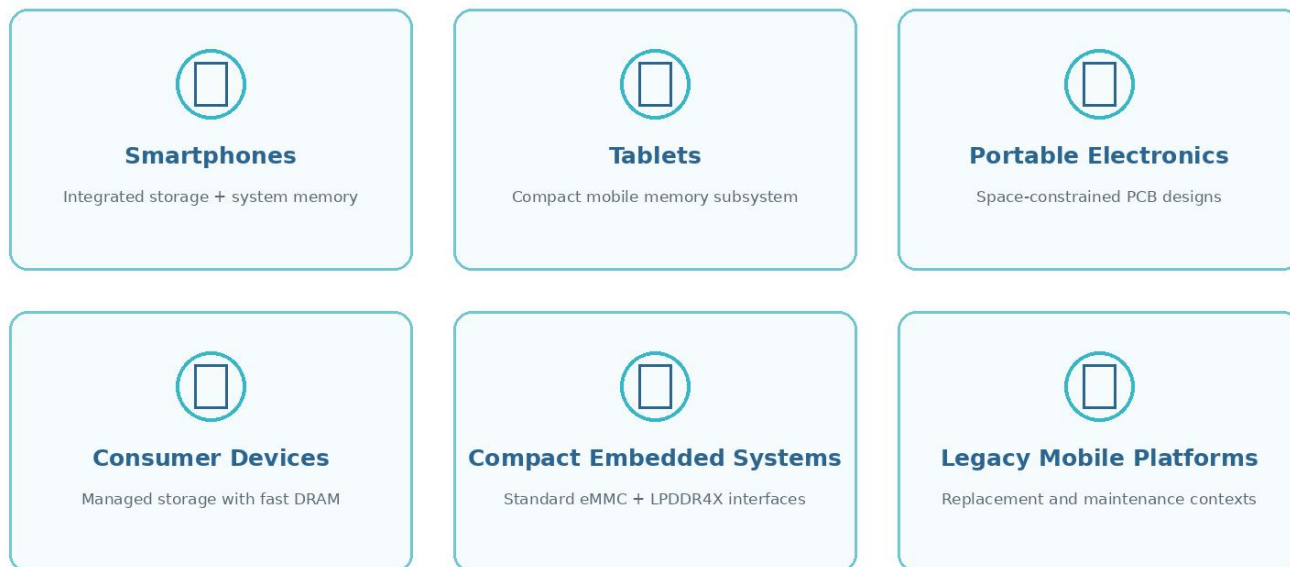
PCB and Assembly Notes

- Use the A1 index and official ball map to confirm component orientation.
- Do not derive the PCB land pattern solely from the simplified illustration.
- Confirm solder-mask definition, reflow profile and inspection criteria with the assembler.
- Use the official manufacturer ball-assignment table for all eMMC, LPDDR4X, power and ground connections.

7. Applications

The H9HP52ACPMAD eMCP solution combines managed storage and high-speed working memory for space-constrained systems where a single-package memory subsystem can simplify PCB layout.

Typical Integration Contexts



These are typical integration contexts for an eMCP device; final design-in requires verification against the exact host, firmware and manufacturer specification.

Application Summary

Application Area	Memory Requirement	H9HP52ACPMAD Advantage
Smartphones	Integrated storage and working memory	64GB eMMC + 4GB LPDDR4X in one package
Tablets / Portable Devices	Compact high-speed memory subsystem	254-ball FBGA with eMMC 5.1 and LPDDR4X
Consumer Electronics	Space-efficient PCB layout	Single-package storage + DRAM integration
Compact Embedded Systems	Managed storage plus volatile memory	Standard eMMC interface with LPDDR4X memory

8. Ordering Information & Disclaimer

Ordering Information

Part Number	Manufacturer	Product Type	Package
H9HP52ACPMADAR-KMM	SK hynix	64GB eMMC + 32Gb (4GB) LPDDR4X eMCP	254-ball FBGA

Product Identification

Item	Description
Content Model	H9HP52ACPMAD
Product Family	Embedded Multi-Chip Package (eMCP)
Memory Configuration	64GB eMMC 5.1 + 32Gb (4GB) LPDDR4X
Target Market	Mobile / Compact Embedded Applications

Revision History

Revision	Date	Description
Rev 1.0	August 2026	Initial SMC technical-reference datasheet release

Disclaimer

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Specifications, features, ordering codes and availability may change without notice. Customers must consult the official manufacturer documentation and verify the complete device marking before final design validation, procurement or production use.

Primary technical basis: SK hynix H9HP52ACPMADAR eMCP Specification, Rev. 1.0, Mar. 2018.

Contact Information

Source Memory Chips (SMC) | Website: www.sourcememorychips.com